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ELECTRICAL SYSTEMS ANALYSIS & DESIGN

WORLD BANK BUILDING – BEIRUT

A Term Project Report Submitted for
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EECE 475: Building Management Systems

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1 Executive Summary

This report details the findings of a field study conducted at the World Bank Building in Beirut. The objective was to analyze the facility's electrical infrastructure, spanning power supply, distribution, protection, backup systems, and building automation. Through a site visit facilitated by the facility management team (MMG), we documented a robust infrastructure designed for high availability, featuring N+1 generator redundancy, hybrid solar-UPS backup, and a Siemens-based Building Management System (BMS). This report presents the technical specifications of these systems, performs critical design calculations (cable sizing, load analysis, and protection sizing), identifies operational challenges regarding utility switching, and proposes targeted improvements for energy efficiency.

2 Project Objectives

The primary goal of this project was to bridge the gap between the theoretical concepts of EECE 475 and real-world electrical engineering applications. Specifically, the team set out to:

- **Adopt and Audit:** Conduct a walkthrough of a major administrative facility to inventory critical electrical systems, including power distribution, backup generation, and automation controls.
- **Analyze and Correlate:** Evaluate the installed equipment against international standards (IEC) and the design principles learned in class, focusing on protection coordination and load management.
- **Validate Design:** Perform independent calculations to verify the sizing of feeders, circuit breakers, and renewable energy components.
- **Optimize:** Identify operational bottlenecks or inefficiencies and propose engineering solutions to enhance power quality and sustainability.

3 Existing System Documentation

3.1 General Building Overview

The World Bank Beirut office functions as a high-performance administrative hub. Due to the critical nature of its operations—including data centers and security systems—the building requires uninterrupted power availability 24/7. However, the facility faces a significant constraint: utility power from Électricité du Liban (EDL) is available for only approximately six hours per day. Consequently, the electrical design treats the utility grid as a secondary source, with the internal generation plant acting as the primary power provider for the majority of the operational day. The entire installation complies with French and IEC electrical standards, ensuring high safety and maintenance protocols.

3.2 Main Electrical Supply and Distribution

The power distribution architecture is designed for resilience. Power enters the facility via the EDL feed and is managed by an Automatic Transfer Switch (ATS) equipped with mechanical interlocks to prevent cross-feeding. This ATS routes power to the Main Distribution Board (MDB), which operates at a standard voltage of 230/400 V (3-Phase, 50 Hz).

Inside the electrical room, the MDB features a robust busbar arrangement labeled L1, L2, L3, and N, distributing power to various sub-main distribution boards located on each floor. The cabling infrastructure utilizes color-coded conductors (Red, Yellow, Blue for phases; Black for Neutral; Green/Yellow for Earth) in compliance with IEC standards. Notably, the main feeders utilize 4-core twisted cables, a design choice that optimizes space within the cable trays and reduces inductive reactance.



Figure 1: External view of the Main Low Voltage Panel.

3.3 Backup Power System (Generators)

To mitigate the chronic lack of utility power, the facility employs a synchronized diesel generator plant. The system consists of three synchronous Volvo generators operating in parallel, with ratings of 200 kVA, 250 kVA, and 130 kVA respectively, providing a total generation capacity of 580 kVA.

Unlike simple standby systems, these generators are managed by a centralized Synchronizing/Totalizing Panel. This advanced control logic monitors the real-time building load and automatically brings specific generators online or offline to match demand. This "load-dependent start/stop" functionality significantly reduces fuel consumption and engine wear compared to running all units continuously. Electrical protection is provided at the source by Molded Case Circuit Breakers (MCCBs), along with integrated overload relays and mechanical dampers for engine safety.



Figure 2: The synchronized generator plant featuring three Volvo units.



Figure 3: The Totalizing Panel responsible for synchronization and load sharing logic.

3.4 Hybrid Solar PV and UPS System

To further enhance reliability and sustainability, the building integrates a hybrid renewable energy system. A roof-mounted solar array, consisting of 39 Q.PEAK DUO panels (410W each), feeds a 10 kW SMA inverter. Rather than just offsetting grid consumption, this system charges a TESVOLT Lithium-ion battery rack located in the UPS room.

For zero-interruption power, a central 60 kVA Uninterruptible Power Supply (UPS) conditions power for sensitive loads such as servers, networking gear, and emergency lighting. This ensures that even during the few seconds it takes for the generators to start and synchronize, critical data and security operations remain unaffected.



Figure 4: Rooftop Solar PV array.



Figure 5: TESVOLT Lithium-Ion battery rack.

3.5 Building Management System (BMS)

The "brain" of the facility is a comprehensive Siemens-based Building Management System (BMS). Physically, the system is decentralized, with BMS panels located on every other floor to gather local data. These panels house Siemens PXC series controllers, network gateways, and I/O modules that interface with field sensors and actuators.

Software-wise, a central dashboard allows facility managers to monitor the entire electrical ecosystem in real-time. This includes granular data on generator performance (voltage, frequency, power factor), UPS battery health, fuel levels, and power transfer status. Beyond power, the BMS actively manages the HVAC system, adjusting fresh air intake based on CO level readings, and monitors the status of the three passenger elevators.



Figure 6: The central BMS dashboard providing real-time monitoring of the generator plant.

3.6 HVAC and Mechanical Systems

- **Cooling:** LG Multi V 5 VRF systems located on the roof.
- **Ventilation:** Fresh air intake and exhaust fans are controlled via the BMS based on air quality sensors.
- **Vertical Transport:** Three Mitsubishi passenger lifts and one car lift controlled by a Variable Frequency Drive (VFD) for efficiency.

3.7 Scope Limitations Security Constraints

It is important to note that the World Bank operates under strict international security protocols. Consequently, access to specific sensitive infrastructure was restricted during our site visit.

- **Restricted Systems:** We were unable to inspect or document the Fire Alarm System, Gas Suppression Systems (for the server room), and the Physical Security/CCTV infrastructure.
- **Impact:** This report focuses exclusively on Power Distribution, Backup Generation, Renewables, and HVAC control. Security and Life Safety systems are excluded from this analysis.

4 Engineering Calculations & Analysis

In this section, we validate the design choices made by the engineering team by performing independent calculations based on the data collected during the site visit.

4.1 Load Analysis & Generator Redundancy Verification

During our visit, the BMS indicated a peak summer load of approximately 140 kW and a winter load of 90-120 kW. Comparing this to the installed generation capacity allows us to analyze the redundancy strategy.

- **Total Generator Capacity:** 580 kVA.
- **Available Real Power:** Assuming a standard power factor of 0.8:

$$P_{available} = S_{total} \times PF = 580 \text{ kVA} \times 0.8 = 464 \text{ kW} \quad (1)$$

Analysis: The available capacity (464 kW) is more than triple the peak load (140 kW). This confirms that the system is designed with **N+1 Redundancy**. Even if the largest generator (250 kVA) is offline for maintenance or failure, the remaining capacity (330 kVA $\approx$ 264 kW) is still sufficient to carry the entire building load with a healthy safety margin for motor inrush currents.

4.2 Feeder Cable Sizing Validation

We will verify the cable sizing for the **250 kVA** generator feeder. **Assumption:** 3-Phase 400V system, Power Factor = 0.8.

Step 1: Calculate Full Load Current (I_b)

$$I_b = \frac{S}{\sqrt{3} \times V_L} = \frac{250,000}{\sqrt{3} \times 400} \approx 361 \text{ A} \quad (2)$$

(Note: Based on draft calculations using conservative voltage estimates, design current is approx 453 A. We will use the conservative figure for safety).

Step 2: Apply Derating Factors

- Temperature ($T_{max} = 45^\circ\text{C}$): $K_1 = 0.79$
- Grouping (Bundled cables): $K_2 = 0.80$
- Total Derating Factor: $K = 0.79 \times 0.80 \approx 0.63$

Step 3: Determine Required Ampacity (I_z)

$$I_z = \frac{I_b}{K} = \frac{453}{0.63} \approx 719 \text{ A} \quad (3)$$

Selection: Based on standard IEC tables for XLPE cables in free air, a **400 mm²** cable (or parallel equivalent) is required to safely handle the current. This matches the installed infrastructure.

4.3 Protection Coordination (MCCB Sizing)

The protection device must handle the full load plus a safety margin while protecting the cable. **Safety Margin:** 1.25 (125%)

$$I_{breaker} = 1.25 \times I_{full_load} = 1.25 \times 453 = 566.25 \text{ A} \quad (4)$$

Result: A standard MCCB frame size of **630 A** with an adjustable trip unit set to approx 600 A is the appropriate choice.

4.4 Solar PV System Contribution

We analyzed the impact of the renewable energy system on the building's total consumption.

- **Total Installed Capacity:** 39 panels $\times$ 410 W = 15.99 kWp.
- **System Efficiency:** $\eta_{sys} = 0.765$ (Combined inverter and cable efficiency).
- **Effective AC Power:** 16 kW $\times$ 0.765 = 12.24 kW.

Daily Energy Calculation: Assuming 5 Peak Sun Hours (PSH):

$$E_{day} = 12.24 \text{ kW} \times 5 \text{ h} = 61.2 \text{ kWh/day} \quad (5)$$

Coverage Analysis: Daily building consumption (120 kW $\times$ 10 h = 1200 kWh):

$$\text{Coverage} = \frac{61.2}{1200} \approx 5.1\% \quad (6)$$

Analysis: The solar system contributes approximately 5.1% of the daily energy needs. This confirms it acts as a supplementary source rather than a primary provider.

4.5 Solar Inverter Sizing

- **Critical Load to Serve:** 6 kW (estimated).
- **Safety Factor:** 1.5.

$$P_{inverter} = \frac{P_{load} \times 1.5}{\eta_{inv}} = \frac{6 \times 1.5}{0.9} = 10 \text{ kW} \quad (7)$$

Result: The installed **10 kW SMA Inverter** is correctly sized.

4.6 Lighting Design Calculations

Design Parameters:

- **Room Dimensions:** $L = 8 \text{ m}, W = 5 \text{ m}, H = 3 \text{ m}$.
- **Work Plane Height:** $h_w = 0.8 \text{ m}$.
- **Mounting Height (H_m):** $H_m = 3 - 0.8 = 2.2 \text{ m}$.

- **Target Illuminance (E):** 300 – 500 lux (Standard Office).
- **Luminaire:** 100 W LED (100 lm/W $\rightarrow$ 10,000 lumens).
- **Coefficients:** Utilization Factor (CU) = 0.8, Light Loss Factor (LLF) = 0.9.

Calculations:

1. Effective Lumens per Lamp:

$$\Phi_{eff} = \Phi_{lamp} \times CU \times LLF = 10,000 \times 0.8 \times 0.9 = 7,200 \text{ lumens}$$

2. Illuminance per Lamp:

$$E_1 = \frac{\Phi_{eff}}{Area} = \frac{7,200}{40} = 180 \text{ lux}$$

3. Required Number of Luminaires:

- 1 Lamp $\rightarrow$ 180 lux (Insufficient)
- 2 Lamps $\rightarrow$ 360 lux (Acceptable)
- 3 Lamps $\rightarrow$ 540 lux (Good)

Conclusion: Install 2 to 3 luminaires to achieve the standard office lighting level of 300–500 lux.

4.7 Elevator Motor Analysis

Equipment: Mitsubishi Elevator Motor (380V, 3-Phase).



Figure 7: Car Elevator controlled by the Mitsubishi Elevator Motor

Motor Specifications (from Nameplate):

- **Type:** 3-Phase Induction Motor
- **Voltage:** 380 V
- **Estimated Power:** 40 HP (based on typical elevator duty)
- **Efficiency (η):** 0.9
- **Power Factor (pf):** 0.85

1. Full Load Current Calculation (I_{FL}):

$$P_{out} = 40 \text{ HP} \times 0.746 = 29.84 \text{ kW}$$

$$P_{in} = \frac{P_{out}}{\eta} = \frac{29.84}{0.9} \approx 33.2 \text{ kW}$$

$$I_{FL} = \frac{P_{in}}{\sqrt{3} \times V \times pf} = \frac{33,200}{\sqrt{3} \times 380 \times 0.85} \approx \mathbf{59 \text{ A}}$$

2. Cable Sizing: The cable is sized for continuous thermal current, not inrush.

$$I_{cable.min} = \frac{I_{FL}}{Derating_Factor} = \frac{59}{0.79} \approx 75 \text{ A}$$

Selection: From standard tables, a 4-core 16 mm² copper cable ($I_z \approx 80 \text{ A}$) is suitable.

3. Circuit Breaker Sizing (Checking Inrush):

- **Starting Current (I_{start}):** For Direct-On-Line (DOL), $I_{start} \approx 6 \times I_{FL} = 6 \times 59 = 354 \text{ A}$.
- **Breaker Selection:** Standard rating $I_n = 63 \text{ A}$.
- **Trip Curve Verification:**
 - C-Curve Trip Range: $5 - 10 \times I_n$.
 - Ratio: $\frac{I_{start}}{I_n} = \frac{354}{63} \approx 5.6$.

Since $5.6 \times I_n$ is at the lower end of the C-curve instantaneous trip range, there is a risk of nuisance tripping.

Conclusion: Use a D-Curve MCB (Trip range $10 - 20 \times I_n$) rated at 63 A to safely handle the inrush current without tripping.

4.8 UPS Battery Sizing

Goal: Size the battery rack for the 60 kVA UPS (Assume 30 minutes autonomy).

Space reserved for Ampere-Hour calculations.

5 Observations & Operational Challenges

Our interaction with the facility management team highlighted practical challenges that go beyond the static design:

1. **Grid Switchover Instability:** The most frequent operational issue reported is the instability during the switchover from EDL to the internal network. Minor faults and voltage transients frequently occur at the moment of transfer.
2. **High Fuel Dependency:** Despite the solar installation, the reliance on diesel generators for 75% of the operational day imposes high operational costs and a significant carbon footprint.
3. **Advanced Motor Control:** On a positive note, the use of Variable Frequency Drives (VFD) for the car lift demonstrates modern design. This reduces mechanical stress and eliminates massive inrush currents.

6 Proposed Improvements

Based on our analysis of the building's specific operational profile—,pecifically its status as a sealed facility relying heavily on generator power,we propose the following targeted improvements:

6.1 Harmonic Mitigation (Active Filtering)

Observation: The building features non-openable windows, necessitating 24/7 mechanical ventilation and cooling. The installed HVAC systems (LG VRF units) and the car lift utilize Variable Frequency Drives (VFDs) and inverters.

Problem: These devices are **Non-Linear Loads**. They draw current in pulses rather than a smooth sine wave, injecting significant **Total Harmonic Distortion (THD)** back into the electrical network.

- **Risk:** Harmonics cause overheating in neutral conductors and transformers. Crucially, **generators** are far more sensitive to harmonics than the utility grid. High THD can cause generator voltage instability and overheating, which is a major risk given the facility runs on generators for 75% of the day.

Solution: Install **Active Harmonic Filters (AHF)** at the Main Distribution Board (MDB) or specifically at the HVAC sub-panels.

Benefit: This will "clean" the power quality, reducing thermal stress on the generators and cables, extending the lifespan of the equipment, and preventing potential corruption of sensitive IT data.

6.2 Solar Expansion Strategy

Observation: The current solar PV system provides only $\approx 5\%$ of the building's energy needs.

Problem: The facility incurs high operational costs and carbon emissions due to heavy reliance on diesel generators.

Solution: Utilize remaining roof space to expand the array. Doubling the capacity to **20 kWp** would allow the system to:

1. Cover the battery charging requirements entirely via solar.
2. Support critical IT loads during daylight hours directly.
3. Reduce the mechanical load on the diesel generators, saving fuel.

7 Conclusion

The World Bank Beirut field project provided an invaluable opportunity to witness the practical application of building management systems. The facility exhibits a high standard of electrical engineering, particularly in its redundancy planning and BMS integration. Our analysis confirms that the power systems are conservatively sized and safe. However, the heavy reliance on diesel generation and the noted grid instability present clear opportunities for optimization. By implementing the proposed smart controls and expanding the renewable energy footprint, the facility can transition from a "robust" system to a truly "efficient" and "sustainable" one.

8 References

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